Chapter 1. Introduction to Nonparametric Statistics

Nonparametric models, parametric inference review, why nonparametric methods, robustness

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1.1 Parametric and nonparametric models

A statistical model is a family $\mathcal{F}$ of distributions assumed to contain the population. It is **parametric** if it is indexed by finitely many real numbers, e.g. the normal family

$$\mathcal{F}_G = \{N(\mu, \sigma^2) : \mu \in \mathbb{R},\ \sigma > 0\}. \tag{1.1}$$

It is **nonparametric** if no finite-dimensional parametrization exists. Typical examples:

$$\mathcal{F}_1 = \{F : \int x^2\, dF < \infty\}, \qquad \mathcal{F}_{\text{all}} = \{\text{all } F\},$$

$$\mathcal{F}_2 = \{F : F \text{ has a density symmetric about some } \theta\}.$$

Both $\mathcal{F}_G \subset \mathcal{F}_1$ and $\mathcal{F}_G \subset \mathcal{F}_2$. A model with a finite-dimensional part of interest and an infinite-dimensional nuisance part, such as $y_i = \beta_0 + \beta_1 x_i + \epsilon_i$ with $\epsilon_i \overset{\text{i.i.d.}}{\sim} F$ unrestricted, is called **semi-parametric**.

In a nonparametric model a **parameter** is a functional $\theta = \theta(F)$: the mean $\int x\, dF$, the median, a quantile, the density f, or a regression curve $m(x) = E(Y \mid X = x)$. Knowing a few such values never determines F. A procedure whose validity (e.g. type I error) holds for every F in a very large family is called **distribution-free**.

The course uses three tools:

1. **Ranks, signs and permutation tests** (Part I): conditional inference given the observed values, exact under exchangeability.
2. **The bootstrap** (Part II): replace the unknown F by an estimate and compute sampling distributions by simulation.
3. **Smoothing** (Part II): estimate a function $f(x)$ or $m(x)$ by averaging the observations near x.

1.2 Review of parametric inference

Definition (Quantile)

For a distribution function F and $p \in (0,1)$, the p-quantile is $Q(p) = \min\{x : F(x) \geq p\}$. For the standard normal, $z_\alpha := \Phi^{-1}(1-\alpha)$.

Proposition 1.1

Let F be the distribution function of an integer-valued X (e.g. $X \sim B(n,p)$). Then $q = Q(p)$ is the smallest integer with $F(q) \geq p$, and $F(q-1) < p \leq F(q)$. In particular $P(X \leq q) \geq p$ and $P(X \geq q) > 1-p$.

Proof. F is nondecreasing, right-continuous and constant on each $[k, k+1)$, so $\{x : F(x) \geq p\} = [q, \infty)$ with q the smallest integer satisfying $F(q) \geq p$. Minimality gives $F(q-1) < p$. Then $P(X \geq q) = 1 - F(q-1) > 1-p$. ∎

Two reference distributions recur. If $X \sim N(\mu, \sigma^2)$ then $(X-\mu)/\sigma \sim N(0,1)$, so $F_X(x) = \Phi((x-\mu)/\sigma)$. If $X \sim B(n,p)$ (number of successes in n independent trials), then $EX = np$, $\operatorname{Var} X = np(1-p)$ (a sum of n Bernoulli(p) variables), and for large n (roughly $np \geq 10$) X is approximately $N(np, np(1-p))$ by the CLT below.

Estimation. An estimator $\hat{\theta}$ is judged by its bias $E\hat{\theta} - \theta$, standard error $\mathrm{se}(\hat{\theta}) = \sqrt{\mathrm{Var}\,\hat{\theta}}$ and mean squared error

$$\mathrm{MSE}(\hat{\theta}) = E(\hat{\theta} - \theta)^2 = \mathrm{bias}(\hat{\theta})^2 + \mathrm{Var}(\hat{\theta}).$$

Under $\mathcal{F}_G$, $\bar{X}_n$ is the best unbiased estimator of μ, and $T = \sqrt{n}(\bar{X}_n - \mu)/s \sim t_{n-1}$ gives the exact interval $\bar{X}_n \pm t_{\alpha/2,n-1}s/\sqrt{n}$. Outside $\mathcal{F}_G$ these optimality and exactness statements are lost.

Testing. A test of H_0 against H_1 rejects when a statistic T falls in a rejection region chosen so that $P_{H_0}(\text{reject}) \le \alpha$ (the **level**); among such tests we want large **power** $P_{H_1}(\text{reject})$. The **p-value** is the probability under H_0 of a value of T at least as extreme (in the direction of H_1) as the observed one; rejecting when $p \le \alpha$ is the same as using the level-α rejection region. In nonparametric problems the central task is to find a statistic whose **null distribution** does not depend on the unknown F.

Proposition 1.2 (Duality of tests and intervals)

Suppose for each θ_0 there is a level-α test ϕ_{θ_0} of $H_0 : \theta = \theta_0$. Then $C = \{\theta_0 : \phi_{\theta_0} \text{ does not reject}\}$ is a $1-\alpha$ confidence set. Conversely, if C is a $1-\alpha$ confidence set, rejecting $H_0 : \theta = \theta_0$ when $\theta_0 \notin C$ is a level-α test.

Proof. If θ is the true value, $P(\theta \in C) = P(\phi_\theta \text{ does not reject}) \ge 1-\alpha$. Conversely $P_{\theta_0}(\theta_0 \notin C) \le \alpha$. ■

The central limit theorem. Much of classical practice relies on large-sample normality.

Theorem 1.3 (CLT)

If $X_1, \dots, X_n \overset{\text{i.i.d.}}{\sim} F$ with mean μ and variance $0 < \sigma^2 < \infty$, then $\sqrt{n}(\bar{X}_n - \mu)/\sigma \xrightarrow{d} N(0,1)$.

Proof. Let $Y_i = (X_i - \mu)/\sigma$ and $\varphi(t) = Ee^{itY_1}$. Since $EY_1^2 < \infty$, $\varphi(t) = 1 - \frac{t^2}{2} + o(t^2)$. By independence $Ee^{itZ_n} = \varphi(t/\sqrt{n})^n = \left(1 - \frac{t^2}{2n} + o(\frac{1}{n})\right)^n \to e^{-t^2/2}$, the characteristic function of $N(0,1)$; Lévy's continuity theorem concludes. ■

Because of the CLT, t-based procedures are routinely applied to discrete, ordinal or skewed data: what matters is that the *statistic*, not the population, is approximately normal. The CLT says nothing about how large n must be. Convergence is slow for skewed F and fails entirely when $\sigma^2 = \infty$ (e.g. Cauchy or slash populations). This is the first motivation for nonparametric methods.

1.3 Why nonparametric methods

- *Validity under weak assumptions.* Rank, sign and permutation tests control the type I error exactly for every F in a large family, at any sample size.
- *Computation instead of derivation.* Permutation and bootstrap distributions replace formulas that are unknown or valid only asymptotically.
- *Flexibility.* Function-valued targets (f, m) are estimated without assuming a shape.
- *Robustness.* Procedures based on ranks, signs or medians are insensitive to outliers.

Example 1.1 (Effect of one outlier)

Take the 8 values 2.1, 2.4, 2.6, 2.9, 3.0, 3.2, 3.5, 25. With the last value, the mean is 5.59 and the standard deviation 7.86; without it, 2.81 and 0.48. The median is 2.95 with it and 2.9 without. The **median absolute deviation** MAD $= \mathrm{median}_i|x_i - \mathrm{median}_j\, x_j|$ is 0.45

with it and 0.30 without. The mean and standard deviation are dominated by a single point; the median and MAD barely move.

An outlier can be viewed as a draw from a heavy-tailed population. Distribution-free methods keep their properties under heavy tails, so they are typically robust (the converse need not hold).